



## Contest Math 8: Intro to Geometry (27079)

Connor Gordon

**Sunday**

Feb 16, 2025 - Nov 9, 2025

12:45 - 2:45 PM ET

### Homework

**Meeting:** 1 2 3 4 5 **6** 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21  
 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

#### Homework: Meeting 6

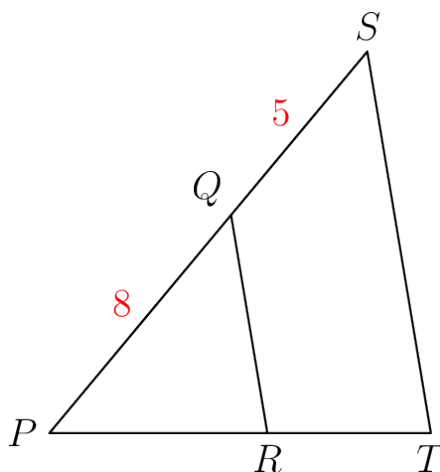
You have completed **0** of **15** challenge problems.Due **Mar 30**.

#### Readings

**Meeting 6:** Sections 5.1-5.4**Meeting 7:** Section 5.5

### Challenge Problems

Problem 1 (10410)

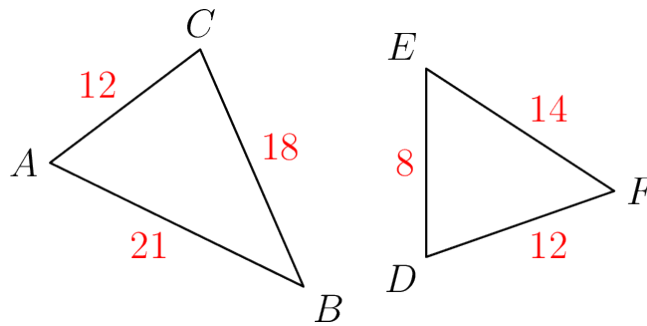
**Problem:**[Report Error](#)In the diagram below, we have  $\overline{QR} \parallel \overline{ST}$ , and  $PT = 10$ . Find  $PR$ .

Your Answer

[Tips](#)

Problem 2 (39033)

**Problem:**[Report Error](#)Which triangle is similar to triangle  $ABC$ ?



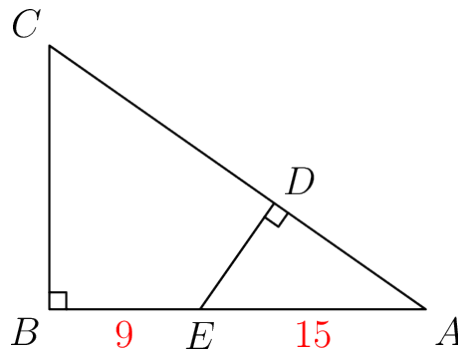
- ☐  $DEF$ 
☐  $DFE$ 
☐  $EDF$ 
☐  $EFD$ 
☐  $FDE$ 
☐  $FED$ 
☐ None of these

### Problem 3 (10411)

#### Problem:

In the diagram, if  $AC = 35$ , then what is  $AD$ ?

[Report Error](#)



Your Answer

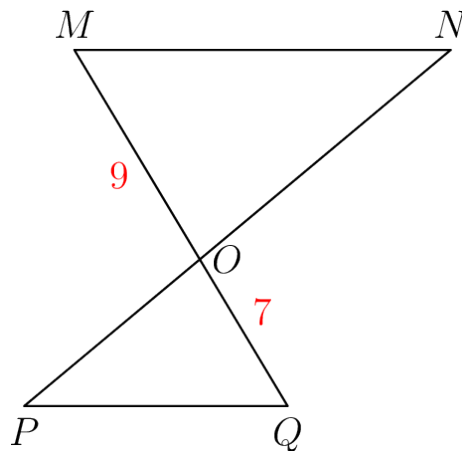
[Tips](#)

### Problem 4 (10409)

#### Problem:

In the diagram below, we have  $\overline{MN} \parallel \overline{PQ}$ , and  $PN = 24$ . Find  $OP$ .

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Your Answer

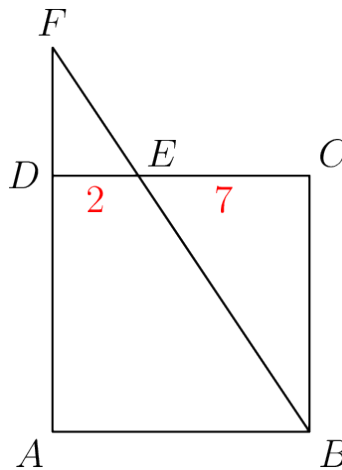
Tips

## Problem 5 (10416)

**Problem:**

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$E$  is a point on side  $\overline{CD}$  of square  $ABCD$ . The extensions of  $\overline{AD}$  and  $\overline{BE}$  intersect at  $F$ . Find  $DF$ .



Your Answer

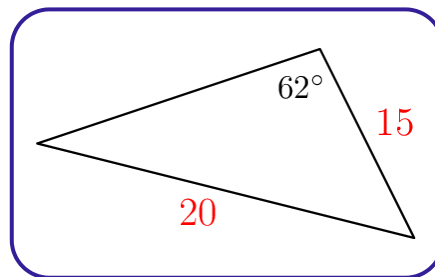
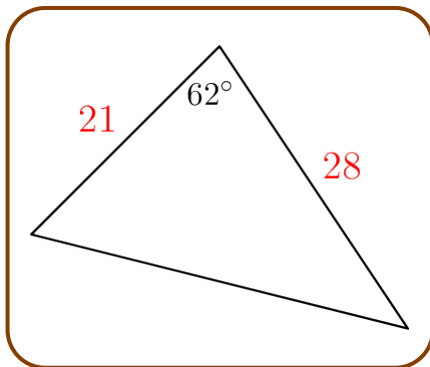
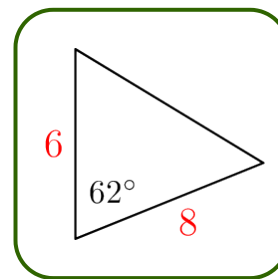
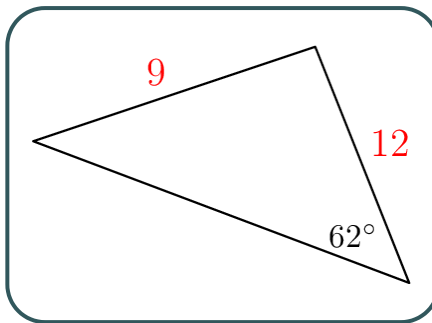
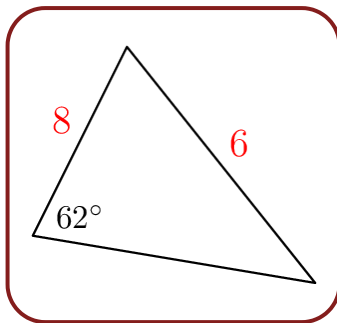
Tips

## Problem 6 (39034)

**Problem:**

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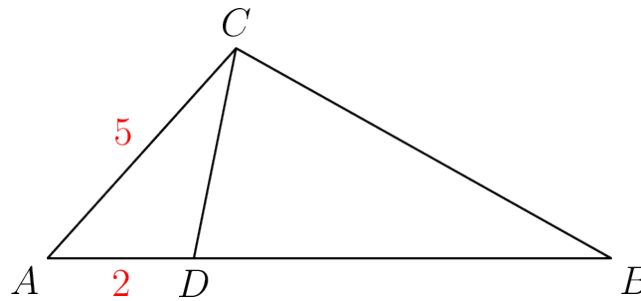
Select the two triangles that must be similar to each other.



## Problem 7 Part (a) (28848)

**Problem:**[Report Error](#)

$D$  is on side  $\overline{AB}$  of triangle  $ABC$ . We know  $\triangle ABC \sim \triangle ACD$ . What is the length  $AB$ ?



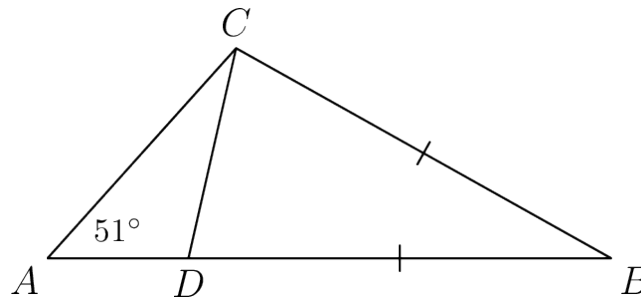
(Based on problem 5.10 in the textbook)

[Tips](#)

Problem 7 Part (b) (23301)

**Problem:**[Report Error](#)

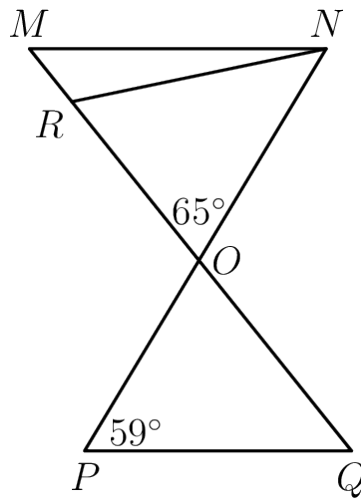
$D$  is on side  $\overline{AB}$  of triangle  $ABC$ . We know  $\triangle ABC \sim \triangle ACD$ . What is  $\angle B$  in degrees?




Problem 8 (23275)

**Problem:**[Report Error](#)

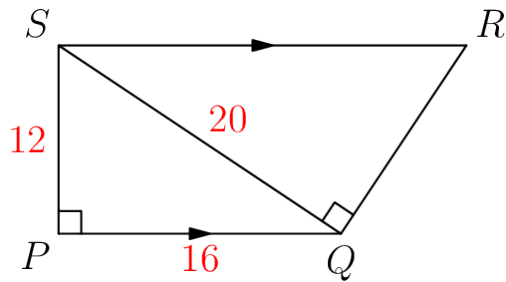
In the diagram below, we have  $\triangle MNO \sim \triangle NRO \sim \triangle QPO$ . What is  $\angle MNR$  in degrees?




### Problem 9 (39039)

#### Problem:

Find  $SR$  and  $QR$ . Enter them in the boxes below, in that order.

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Your Answer:

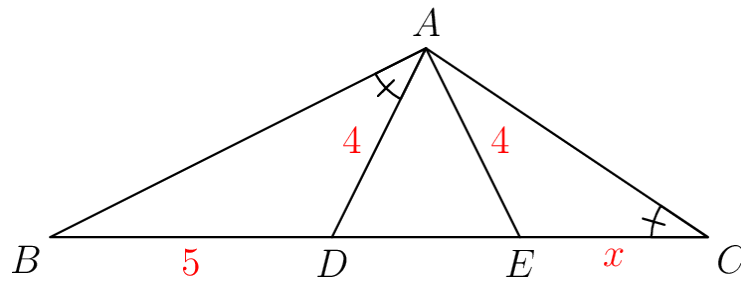


### Problem 10 (39036)

#### Problem:

Find  $\mathcal{X}$ .

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Your Answer

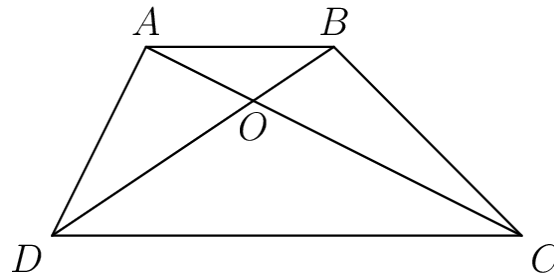
Tips

## Problem 11 (39032)

**Problem:**

In the diagram,  $\overline{AB} \parallel \overline{CD}$ . Select the two quantities that must be equal.

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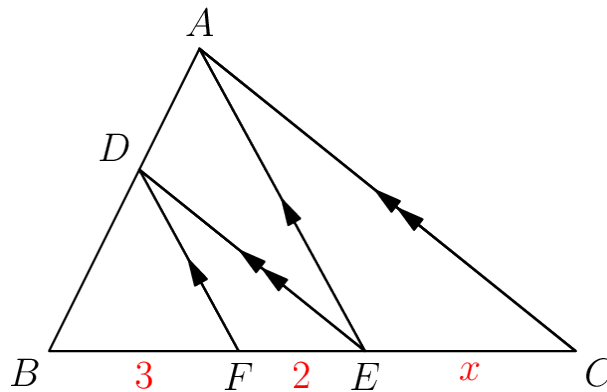
 $AO \cdot BO$  $AO \cdot CO$  $AO \cdot DO$  $BO \cdot CO$  $BO \cdot DO$  $CO \cdot DO$ 

## Problem 12 (39035)

**Problem:**

Find  $x$ .

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Your Answer

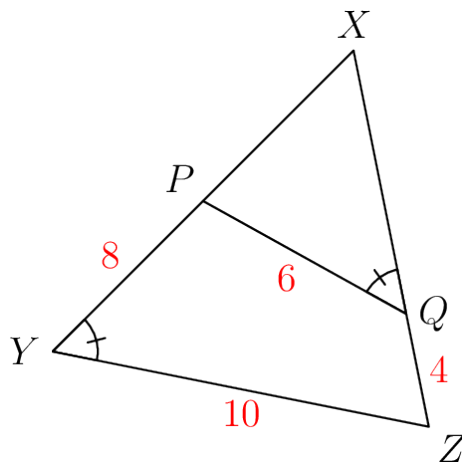
Tips

## Problem 13 (39037)

**Problem:**

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Find the perimeter of triangle  $XYZ$ .



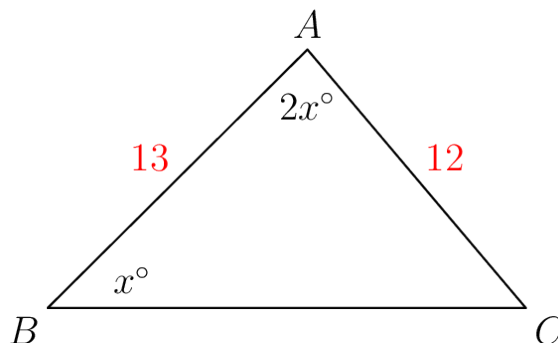
Your Answer

Problem 14 (39038)

**Problem:**

[Report Error](#)

Find length  $BC$ .



**Beast Problem:** 🐉

This is a **Beast** problem. Answering this problem can only increase your score. You will not lose points for skipping this problem or answering incorrectly.

Your Answer

[Tips](#)

Book Problems (39040)

In addition to the homework listed here each week, you also need to complete several additional homework problems from your textbooks. This week's book problems are:

- *Introduction to Geometry*: 5.12, 5.2.5, 5.25, 5.29
- *Art of Problem Solving, Vol. 1*: Exercise 11-17

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